Vault, providing a signal to security teams that something unusual has occurred.

11.5.4 Key derivation

A complementary approach to secure distribution of secrets is to reduce the number of secrets your application needs in the first place. One way to achieve this is to derive cryptographic keys for different purposes from a single master key, using a *key derivation function* (KDF). A KDF takes the master key and a context argument, which is typically a string, and returns one or more new keys as shown in figure 11.9. A different context argument results in completely different keys and each key is indistinguishable from a completely random key to somebody who doesn't know the master key, making them suitable as strong cryptographic keys.

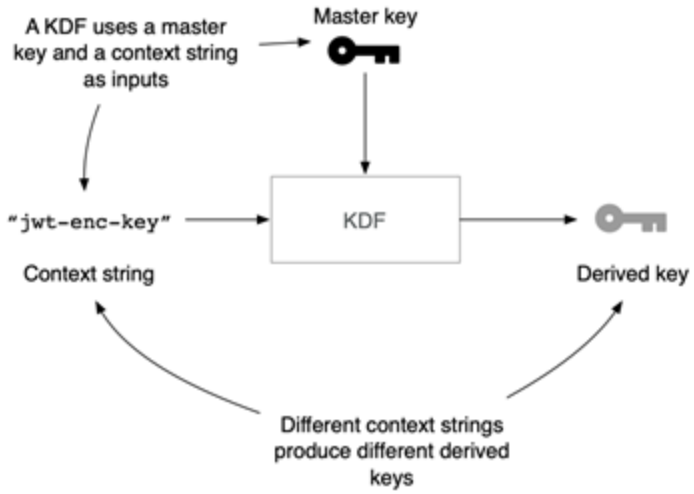


Figure 11.9 A key derivation function (KDF) takes a master key and context string as inputs and produces